

“How Can I Help You?”: User Perceptions of Privacy in Retail Chat Agents

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ABSTRACT

Chatbots that request user data are proliferating in the retail industry. There are studies showing that increase in perceived privacy risks can decrease their adoption rate. In this study, we interview eight participants about their reasons behind these perceived privacy risks so that developers can make retail chatbots more privacy-preserving and respectful of user choices. This short thematic analysis highlights some of the privacy concerns in such bots and some of the recommendations that will be helpful for future work to build on.

CCS CONCEPTS

• **Security and privacy** → **Human and societal aspects of security and privacy; Social aspects of security and privacy; Usability in security and privacy; Privacy protections;** • **Human-centered computing** → **User studies.**

KEYWORDS

chatbots, privacy, conversational privacy, trust

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1 INTRODUCTION AND BACKGROUND

Chatbots have increased online presence due to their integration in a variety of industries, like finance, retail, healthcare, insurance, and customer service. The past few years have seen an explosion in the number of chatbots (or ‘automated chat agents’) being used specifically in the retail space. According to a recent estimate, chatbots will drive \$112 billion in retail sales¹ as more companies see increased financial value in deploying them.

Such interactive bots can be either retrieval-based (generates response based on specific query terms, like [19]) or generative (learns how to respond to diverse input through machine learning

techniques, like [25]), with the majority of retrieval-based chatbots moving towards more automated, generative models. For instance, chatbots engage in more and more human-like conversations [11, 15, 24], like making reservations and starting informal conversations. Chatbots also exist across different modalities, accepting text, voice, and gesture input. As these chat agents proliferate, more research is focusing on factors inhibiting their adoption.

Users are divided about chatbots usage due to usability and trust. One of the reasons why users are still skeptical are privacy issues [1] with chatbots. Prior research has shown that privacy can indeed impact the adoption of chatbots and erode trust [16]. Negative impression of chatbots are driven by two primary privacy factors - perceived risk and familiarity. Perceived risk or “creepiness” was found to be a driving factor for people to move away from financial chatbots [6], healthcare bots [17], customer service chatbots [22], and advertisement bots [14]. Adding a personality profile further enhances this risk with people discussing “*sensitive content that is too embarrassing to ask another human*” [29], and volunteering content with reduced inhibitions.

While less significant that perceived risk, familiarity with specific bots and the organizations which developed such bots also increased trust [6]. However, chatbots across different industry verticals motivate different levels of user privacy perception. As Cardona and colleagues have argued, perceived usefulness overtook perceived risk as deciding factor for chatbot adoption, but for specifically for insurance bots [23]. In addition, there is security anxiety surrounding the implementation of chatbots since malicious bots can be implemented on popular platforms [13].

User perception aside, some messenger-based chatbots do not provide enough information about privacy policies or provide adequate permission settings [10], eroding users’ ability to make privacy choices for themselves. In some generative bots which provide social support, greater information disclosure has been observed due to the emotional bond-forming nature of these bots [21, 27]. Furthermore, previous studies have shown that in financial transactions, users tend to dismiss privacy concerns when provided with the right benefits (like a discount on purchases for example) [5]. Similarly, Lee et al. did not find users mentioning privacy concerns around personal information in cryptocurrency bots [18]. In such cases, transparency about data practices is absolutely necessary, even if not driven by user perception.

There is some incremental progress towards making chatbots more secure and privacy-preserving. Some recent work has been studying different attack surfaces for chatbots [28] and developing initial tool ideas for secure technical controls [10]. One study has also found that increased visibility and user control actually improved chatbot adoption by reducing perceived privacy risk in

¹<https://www.marketingdive.com/news/study-chatbots-to-drive-112b-in-retail-sales-by-2023/554416/>

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healthcare chatbots [8]. Another article mentions a number of ways chatbots can modify their interfaces, in order to ensure compliance with the recent General Data Protection Regulations (GDPR)², which is critical for financial institutions [26].

Nevertheless, though privacy has been an important focus and deemed to impact chatbot adoption, there is less research around what specific privacy risks are perceived by users. For retail chatbots, which we define as bots that can make a financial transaction - like making purchases on behalf of the user, handling payments, adding products, or canceling orders, we want to understand how users control their information and perceive how their information is being used. These AI-based chatbots can provide all of the aforementioned online customer services 24x7. Additionally, we wanted to understand how this privacy perception extends to companies which develop these bots. Our research question is the following: **What are some of the privacy perceptions that users of retail chatbots have?**

Briefly, we found that users have selective data types that they will not share, but they are comfortable interacting with these chatbots as long as they are helpful. However, there are information boundaries that people maintain and privacy concerns that they have while interacting with bots. Furthermore, we report on the risks that users inadvertently perceive during interaction from the bot, and the extent to which they extend these risks as a consumer to the organization implementing these bots.

To keep the scope of this study narrow, we focused on retail chatbots that can conduct financial transactions. Furthermore, we only looked at chatbot experiences that were text-based since modality can affect how users perceive such agents [20]. We recruited eight participants who have used chatbots that meet this criteria and conducted semi-structured interviews on what their privacy concerns were and what factors affected their trust on such retail bots. Given that there is an intuitive push from industries to make their products more trustworthy, we conduct a thematic analysis to highlight some of the major concerns that companies can address for their retail chatbots. This improved privacy would not only improve product quality [3, 12] but also improve user trust [9], giving companies an incentive to attend to these concerns. This paper also contributes to the limited existing research around understanding privacy of chatbots, paving way for future large scope research around different industry-specific chatbots as they increasingly become more intelligent.

2 METHOD

Prior research has shown that perceived privacy risks can cause less adoption in various types of chatbots, but there is less exploration of some of the reasons behind these perceived privacy risks, especially for retail chat agents. Since our goal was to understand what some of these privacy concerns are and how they affect users' understanding of these bots, we decided to conduct semi-structured interviews with a group of participants who were regular retail chatbot users. All study procedures were reviewed and approved by the university's Institutional Review Board (IRB). The details of recruitment and the study session is as follows.

2.1 Recruitment and Screening

We recruited participants during Fall 2021 via flyers posted at the university campus and on social media platforms like Reddit and Twitter. In addition, we posted advertisements to campus mailing lists. The advertisements included a link to an online screening questionnaire. Anyone who encountered the link could fill out the screening questionnaire. Anyone at least 18 years old was eligible to participate. Since the study was conducted during the COVID-19 pandemic [2], we only included participants who could attend the study online in case a socially-distanced in-person setup could not be arranged. We excluded participants who did not meet the eligibility criteria of being 18 years old or had no prior experience of using retail chatbots. This was to scope our study to chatbots with transaction capability.

2.2 Participants

A total of eight people participated in the study. Table 1 provides an overview of participant characteristics. Participants were equally split between men and women. Participant ages ranged from 23 to 49 years. All participants had diverse educational backgrounds and were residents of a university town.

We conducted the study as per participant availability following the pandemic. We recruited eight participants which is considered enough participants for a small group study [7] using thematic analysis. This approach allowed us to interact more freely with each interview participant.

2.3 Study Session

Each study session was conducted with an individual participant. Participants were introduced to the study's purpose by explaining that our goal was to understand how people interact and share information with retail chatbots. This was done to not bias participants towards thinking about privacy. When they would bring up a topic related to their privacy perceptions, the interviewer would ask probing questions to get more context.

The session began with general questions about chatbot usage. These were followed by questions on challenges faced by participants while managing their data and what design changes they would like chatbot platforms to make with respect to their data management. In the screening questionnaire, we had asked participants to note their most extensively used chatbot, and the interviews with each participant directly referred to their experience with their latest used retail chatbot. We did this to get participants thinking about their latest interaction with retail chatbots and draw on that experience. The list of chatbot categories can be found in Table 1.

With permission from the participants, we screen recorded the study sessions and audio recorded the post-study interviews for transcription and analysis. The study lasted about 45–60 minutes per session. We provided participation compensation of U.S. \$10.

2.4 Analysis

We first transcribed the audio recording to transcripts for the first five participants. We began coding using an iterative inductive approach, noting points of frustrations, commonality, and data-preserving approaches. We also separately noted the suggestions participants made in terms of how their data could be protected by

²<https://chatbotsmagazine.com/is-my-chatbot-gdpr-compliant-fc18dd57f9e6>

P. No.	Age	Gender	Major	Chatbot Usage Category
P1	36	Male	Cognitive Science	Social Media
P2	26	Female	Astrophysics	Assistants, Beauty
P3	23	Female	Marketing	Financial, Education, Retail
P4	23	Female	Media Studies	Beauty, Retail, Education
P5	23	Female	Elementary Education	Assistants, Utilities (Cable, Electricity)
P6	29	Male	Computer Science	Food delivery, Financial, Retail
P7	49	Male	Philosophy, Computer Science	Retail, Assistants
P8	25	Male	Optometry	Education, Financial, Retail

Table 1: Participant demographics showing age, gender, educational background, and type of retail chatbot used. The social media chatbot used by P1 can make payments.

chatbots (or indicated by them otherwise). We then analyzed the rest of three transcripts as and when they were transcribed.

Our coding process included open coding, pattern development, and aggregation of responses by themes, following a thematic analysis approach as outlined by [7]. We formed a codebook of higher-level themes synthesized from the eight participant sessions. Coding was done by one author. Codes that appeared only once were excluded from the analysis. We then made a pass through the entire data, aggregating responses by themes across participants. We ended with a total of five themes around data privacy perception and expectations, discussed in Section 3.

2.5 Limitations

The study was based on participants’ self-reported perceptions and behavior. As such, it is difficult to state how they might have approached their interaction with chatbots in a naturalistic setting. We also had a small sample, which while providing rich, personal data, is insufficient to make reproducible, generalized statements about chatbot interactions at large. This study is meant to be an initial exploratory work into understanding how privacy perceptions apply to retail chatbots in general, which has had limited exploration in literature. In future work, we would improve rigor by coding with multiple coders and increasing the number of participants to reach data saturation.

3 FINDINGS

In contrast to [4], none of the participants in our study mentioned that they do not care about their data, perhaps due to the ability of retail chatbots to conduct transactions on their behalf. On the contrary, participants described privacy challenges from retail chatbots they had encountered as a combination of old and new concerns. They viewed such automated chat agents as an extension of the platform and as contributors to the personal information dossier that such platforms collect. However, they also agreed that the automated nature of these agents create additional privacy intrusions, like automated spam, unnecessary data querying, and integration of third-party bots. The following sections illustrate some of these themes.

3.1 Chatbots Make Targeted Ads and Spam Easier

With the exception of one, participants wondered if some chatbots collect data that users input and then use it along with browser cookies to provide targeted advertisements to users. This made some participants skeptical about sharing information with such chatbots. Even if they were willing to share some basic information that would help the bot locate their account and respond to their query, they did not want to share detailed information that could make them targets for advertisements tailored to their demographics:

I’ll give them basic info, but I don’t really want them to know who I am, what kind of person I am, what I like, what I don’t like. (P5)

One participant, P7, was worried that even such basic information entered through chat agents, like zipcode for example, would lead to targeted advertisements because it was in the companies’ best interests to do so using chatbot data:

Companies like giving you, you know, you go to a website to buy something and based upon your zip code and the other information they figure out, Oh, you’re earning this much. So you, so we’ll give you a higher price than somebody else. So it’s not in my best interests to give them that information. It’s in their best interests to get it but not mine. So I always have to think about that. (P7)

Thus, they always think before giving out information to retail chatbots which can be used later for targeted solicitation. The non-transparent nature of these chatbots make it difficult for users to understand how their data is being used, similar to messenger bots [10]. When asked, none of the participants knew if there was a separate privacy policy for chatbots that outlined whether their data was being used to deliver a service or repurposed for others, like advertisements. This made them distrusting of these bots, and by extension, the companies who develop them.

About half participants thought that chatbots make it easy for companies to mine user data. According to P1, it would have been expensive otherwise, if they had to hire a person to call people and collect information. However, due to automated chatbots that retail outlets use, it is easier for them to call people and the turnaround for spam was thought to be higher by P1.

what kind of data that they're collecting and how much you like is and how much is being sent to you, how much are they taking for themselves to be sold on whatever - it's just like a data mining operation. (P7)

According to P7 above, the whole situation was a "data mining operation" where companies would just try to mine people's information by using voice-based retail chatbots.

3.2 Retail Bots Aggregate Non-Removable Personal Information

All participants found "friendly" chatbots to be intrusive and not very helpful. One participant mentioned how they did not like the chatbots that pop-up immediately when they enter a website, asking for more information. As P3 mentioned:

"Oh how can I help you?", because I feel like those are not really helpful. They are just like really a way to get my e-mail, or more personal information. (P3)

Friendly chatbots that tried to engage users in conversation on a retail platform were seen as collectors of personal information. Often, these chatbots would ask for information to be filled out which one participant found not only tiresome and time consuming due to its lengthy nature. In some instances, these chatbots were gathering a lot of unnecessary information about her that were not relevant to the task she was trying to accomplish:

[Some chatbots are like] "before we answer your question, can you tell us a bit about yourself? And then they would have four or five fields of information filled out - a lot of times. That was honestly just for time purposes. I didn't want to go through the process of filling it out when I knew that I could just call or I could try to navigate the website more. But then there is the matter of not being very comfortable giving a lot of personal information and filling those fields out (P5)

This made participants feel that either companies that operate such bots are just harvesting more personal information than they need and wasting consumers' time. More importantly, three participants also observed that they were unable to request deletion of the chats that they had provided to the chatbot. They believed these chats were being aggregated for indefinite periods of time. They wanted to be able to request deletion like they can for text messages or otherwise have a fixed retention period after which chats "expire".

However, even though they did not appreciate data mining using chatbots, most participants agreed (six of eight) that chatbots already have information that they have collected through other channels. For example, if a user was logged into Amazon, then the Amazon Returns chatbot would already have their information that they had provided while setting up the account. Similarly, if a user was enquiring about a purchase and provided an order number, the chatbot (like any customer service personnel) could look up associated personal information like name and home address of the recipient. Thus, while all participants recognized some forms of data as sensitive, they were skeptical if they could prevent access to

some information that chatbots already have on them. According to P2:

Social Security card [...] can be very personal. But now I gave my name, I would give my e-mail. I don't think I'll guess - I would say I wouldn't have given my phone number, but I think I put that in my order anyway too, you know. Oh. That's hard. I don't know if I have... I say things like, I don't think I put that in there, but you put your birthday, you put all these things and when you order stuff anyway (P2)

Participants believed that different pieces of information provided to the bot and the company that developed the bot could be linked together to create a profile.

3.3 Types of Information (Not) Shared

Seven of eight participants mentioned specific information types that they considered sensitive and would not share with a bot. This included addresses, dates of birth, Social Security numbers, phone numbers, and so on. Two participants, P1 and P4 mentioned financial information like bank details and credit card numbers as information that they would never share with a chatbot. P1 also believed that the only way to cause any actual harm would be to get access to financial information. Without it, companies (or malicious actors who compromise bot data from these companies' servers) would not be able to do too much:

Like as long as I'm not giving them my credit card number or any personal information, they can record all the information they want. And I don't think it's going to be very useful. (P1)

Conversely, P7 believed that not providing such information would make it harder for companies to collect information about users, but they would eventually get it. However, P7 refrained from providing any sensitive information and it making it easier for companies to reach that level of aggregation.

3.4 Privacy Perceptions Extend to Companies Who Develop Them

When asked how chatbots deployed by companies affected their perception about these companies, participants took a layered approach to the answer. For example, P1 mentioned that he thought user-created chatbots are much safer because they are well-intentioned. Since they are created by a single user, their goal would not be malicious, especially in closed group settings where manual content moderation is possible. P1 cited the example of Reddit and Twitch bots, where authorized administrators or users create automated chatbots to answer questions posed by viewers or to moderate chats. Interesting to note is that single users can also create malicious bots, but when thinking of bots built by a single developer in their trust group, some participants were more likely to be more trusting of them.

However, it was different when it came to chatbots that were developed by large companies. According to P6, large companies were more secure when building chatbot systems than smaller businesses because they have allotted a significant amount towards security:

It will depend on, I guess, if the company is mature enough, does it, do I perceive that the organization as having a sound technology? How mature is it in terms of security? Because I may, well give my card details or basic details to something like an Amazon rather than a startup, which I have not heard of or maybe I'm hearing for the first time - [where] I personally didn't happen to know the reputation of the company. And secondly, I don't know how it's going to handle the data. But for something like Amazon, I guess it's just that the trust that they have to [maintain in] the society, I guess. (P6)

Participants trusted companies with greater resources to keep the information they provide to bots secure, because they are socially accountable. They expect internal policies in such companies and law enforcement to audit what personal information bots developed by these companies collect.

3.5 Perception of Third-party Integration in and with Chatbots

Perceived risks from chatbots is a driver of low adoption [6], especially for third-party integrations. Diving deep into reasons for perceived risk, participants mentioned two areas of concerns. First, they were concerned about chatbots that collect information and share it with other interested parties. In this case, one participant believed that it would provide multiple parties with access to their data. They wanted companies to limit the data they share with third-parties when the data came from their conversation logs with a retail chatbot.

The second area of concern was that third-party chatbots that are integrated into legitimate platforms, without heed for data privacy. As P8 mentioned, task-based retail chatbots might be developed by an external company and integrated into a different platform. In such cases, it is hard to keep track of the data trail. Often, companies do not inform existing users before deploying such chatbots. According to P8:

[Third-party chatbots] could potentially abuse - just looking out for the chatbots like [the university's], their website actually has a chat bot that's outsourced to something called KB or something. I think that's their branding for it. But it's it's branded [by the university]. I don't know. I just remember going on [to] use website and all of a sudden there was this chat bot and I was like "this is not made by [the university]". (P8)

Participants viewed this lack of information as breach of user trust. They expected chat-like conversations to be private between them and the the bot (and the company developing the chatbot). They did not want their chat information to be transferred or sold to other parties without their knowledge or permission. This concern also extended to third-party bots which were not developed by a company internally but outsourced to another party.

4 DISCUSSION AROUND POSSIBLE PRIVACY-PRESERVING RECOMMENDATIONS

In this study, we observed that privacy concerns around user-bot interactions in a retail setting were unique. Users expected privacy protections similar to what they would expect in closed chats. However, they were aware that such protections were not in place - it was another source of data for the bot's parent organization. They instead believed that the nature of these help bots make it easier to aggregate data and target advertisements at consumers. They wanted measures that would protect their chat data from extending beyond the chat through better transparency, less tracking, more relevant data collection, and less third-party sharing.

- *Develop Consent Renewal System for Tracking:* Participants in this study found spam and targeted advertisements based on chatbot input frustrating (Section 3.1). While talking about design ideas, Participant P7 recommended that the default setting should be to not track users based on their chat input. P7 further mentioned that even if consent has been provided by a user for sending users advertisements, they should have the opportunity to opt-out if they want. Such a consent renewal system would also keep marketing lists updated on the company's end.
- *Increase Relevant Data Collection:* The nature of retail transactions often require bots on retail platforms to ask users for several pieces of information. However, as we found in our study, sometimes such information can be irrelevant to the query as perceived by the user. Sometimes, these chat agents might request information that the user does not want to provide to an automated system (Section 3.3). Four participants wanted chatbots to minimize the data collected and not other unnecessary information that are unrelated to the their queries.
- *Reduce Third-party Sharing:* Participants wanted their conversations with respective retail chatbots to remain private. They did not want it to be shared to other parties unless explicitly asked. This was especially true for platforms which integrate bots at a later point in time, and outsource the development to another party (Section 3.5). In such cases, the user might trust a platform, but not the company developing the bot and deploying it on the platform, similar to Lau et al [16]. In such cases, there should be policies around who receives the information from users and how the information is used.
- *Improve Chat Transparency:* A common thread throughout our discussion with participants was that they were unsure of what data is being collected and for what purpose, similar to [10]. This was especially true when they discussed retail bots aggregating personal information (Section 3.2) or typed of information they would not share (Section 3.3) because the purpose of collection was unclear to them. Participant P1 recommended that when retail chatbots are aggregating information about them, they should provide disclosure messages like "your input is being recorded", similar to customer service calls and users should be given the opportunity to see data that the chatbot stores about them.

While these are some of the possible avenues bot developers should think about, they are in no way a complete list of recommendations on how privacy-sensitive chatbots can be designed. Every chat agent platform is unique in the type of data collected and how it is handled. To create a comprehensive set of design guidelines would require further research and testing, which we plan to do as part of future work.

5 CONCLUSION

Our findings in this study highlight some of the expectations of privacy in conversational interfaces for human-bot conversations in a retail setting. Supported by these findings and suggestions by participants, we discuss some possible avenues of privacy-preserving changes that developers of retail chatbots can make to align their product to users' expectations. These guidelines would also improve such interfaces to become more trustworthy and conscious of the privacy choices of this participant group. While further research is required before we can develop a complete set of privacy guidelines for chatbots, this user-driven approach in developing privacy controls will ensure that privacy remains a fundamental goal in the design process.

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